King Saud University
College of Computer and Information Sciences
Department of Information Technology

IT 320 Course Project
Semester-2, 1447H

Software Product Release
< KSU سُبل>

Prepared by

<Dana Alotaibi, 445203123>
<Gharsah Almusaeed, 445200356>
<Razan Alkahtani, 445202333>
<Ghena Binjadid, 445202185>

Supervised by
<Ms.Ghaida Alfayez>

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Chapter 1: Introduction

A web-based transportation reservations management system for university students play a vital role in ensuring punctuality, safety, and accessibility to education. At King Saud University, many female students rely on buses commuting from nearby metro station to campus. However, the absence of a centralized digital system for managing bus capacity and student access has led to overcrowding, uncertainty about seat availability, and inefficient use of transportation resources.

The system provides students with the ability to search for available buses, view real-time seat availability, reserve seats, and cancel bookings when necessary. It also offers an optional scheduling feature that allows administrators to allocate bus arrival times on a daily, weekly, or monthly basis, helping students anticipate transportation availability in advance and reducing uncertainty and stress related to daily commuting.

The importance of this technological solution lies in improving commuting efficiency, reducing congestion, and enhancing student experience through fair seat allocation and reliable scheduling. On a broader scale, smart transportation systems contribute to sustainable urban mobility and better utilization of public resources.

The proposed solution will positively impact transportation management both locally and globally. Locally, it will improve the daily commuting experience of female students at King Saud University by providing organized access to buses, clearer visibility of available seating, and predictable transportation arrangements, helping reduce stress and improve punctuality. It will also support campus transportation operations through centralized data and structured scheduling. Globally, this type of system reflects a scalable approach to smart campus mobility, offering a model that can be adopted by universities and institutions worldwide to enhance public transport coordination, optimize vehicle utilization, and support sustainable urban transportation practices.

This document presents the project background, defines the transportation problem in detail, describes the proposed solution, and outlines the product vision guiding system development.

1.1 Problem:

A significant real-world problem faced by female students commuting to King Saud University is severe crowding and insufficient seating availability on transportation buses operating between metro station and campus. During peak hours, the number of students often exceeds the available seats, forcing many to stand or wait while buses become excessively crowded. This situation is worsened by the absence of a structured system to manage student distribution across buses, leaving students uncertain about seat availability and contributing to daily stress, discomfort, and inconsistent access to transportation.

In practice, when a bus arrives, it is required to wait for a fixed period to load students, even if it becomes fully crowded before that time ends. During this process, the following bus must also wait until the first bus departs, regardless of whether it already has available capacity. This creates unnecessary delays, increases congestion at the station, and causes inefficient use of buses, as some vehicles remain idle while others are overloaded. As a result, students experience stress, uncertainty, and extended waiting times, especially during peak hours.

This project focuses specifically on the overcrowding of buses, the absence of visibility into available seating, and the lack of centralized management of student distribution across vehicles. These issues result in uneven bus utilization, with some buses becoming excessively crowded while others remain underused, creating confusion among students and limiting fair access to transportation, particularly during high demand hours.

1.2 Solution:

The proposed solution is a web-based system that manages transportation buses between metro station and King Saud University. The platform supports two user roles: Administrator and Student.

The Administrator is responsible for managing all system data, including adding new buses with details such as bus number, driver name, and capacity, and editing existing records to update operational status. When a bus is no longer available, the system allows the Administrator to remove it from active service. Deletion is handled through a controlled process where the bus is marked as inactive and automatically hidden from student search and reservation views, ensuring that historical records remain stored while preventing unavailable buses from appearing in schedules.

Students can register and log in to the system to search for trips that allow students to quickly locate available buses based on real-time seat capacity, view real-time seat availability, reserve seats, and cancel bookings when necessary. In addition, the platform supports an optional scheduling function that enables administrators to assign bus arrival times on a daily, weekly, or monthly basis. This functionality allows students to plan their commute ahead of time, promotes smoother distribution of students across buses, and minimizes unexpected crowding.

By centralizing fleet information and student reservations in one platform, the system improves transportation organization, reduces congestion, and ensures fair access to seats, resulting in a more reliable and structured commuting experience.

1.3 The Product

1.3.1 Product Vision

For a_______________ (Female student at King Saud University)

Who _______________ (faces daily overcrowding and uncertainty when using transportation buses.)

The__________ (KSU سُبل) is a _______(web-based transportation reservations management system).

That_______________ (provides real-time seat availability, structured reservations, and optional scheduling of bus arrival times on a daily, weekly, or monthly basis).

Unlike_______________ (Darb).

Our product ______________(introduces fixed capacity management for each bus and supports flexible reservation options on a daily, weekly, or monthly basis, ensuring fair seat allocation, reducing congestion, and optimizing the distribution of passengers across available vehicles).

1.3.2 Product Roadmap

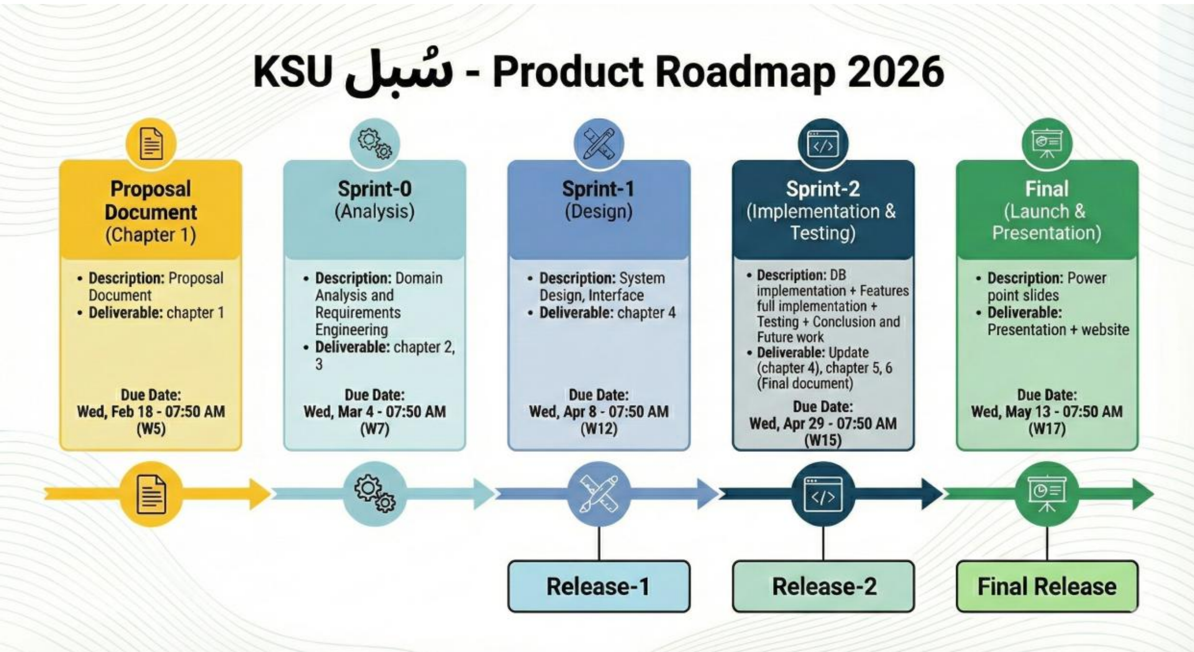

Figure 1: KSU سُبل Project Roadmap (Generated using Google Gemini) [1]

1.3.3 Objectives

1.3.3.1 Product Objectives (Customer Focus – Value)

- Provide a web-based system for female students to view available buses and access transportation information in real time.
- Enable students to register by creating an account using a unique username, and a password.
- Enable students to log in securely using their registered username and password.
- Enable students to log out to ensure account security and privacy.
- Enable students to search for available buses by time and current seat availability.
- Enable students to view their resrvations.
- Enable students to edit their reservation within 2 minutes.
- Enable students to reserve seats based on time and seat availability.
- Enable students to cancel reservations when necessary to free seats for other students.
- Allow administrators to add new buses by entering bus number and bus capacity information.
- Allow administrators to edit existing bus details, such as capacity, and status (available, full) to keep records updated.
- Allow administrators to delete buses that are no longer in service by removing them from the active schedule while preserving system records.
- Support an extra functionality that allows the student to allocate bus arrival times on a daily, weekly, or monthly basis to help them plan ahead and reduce congestion.

1.3.3.2 Project Objectives (Solution Focus – Plan)

- Analyzing user needs by identifying challenges faced by students in current transportation processes.
- Designing system requirements, including user roles, database structure, and functional features.
- Developing the web application interface for both students and administrators.
- Implementing core functionalities such as authentication, bus management (add, edit, delete), search, reservation handling, and scheduling.
- Building and integrating the database to store buses, users, reservations, and schedules.
- Testing system functionality to ensure reliability, usability, and data accuracy.
- Deploying the system and preparing documentation describing system usage and architecture.

1.3.3.3 Learning Objectives (Student Focus)

- Gain hands-on experience in full-stack web development.
- Improve understanding of software engineering practices, including requirements analysis, system design, and implementation.
- Learn how to manage databases and integrate them with web applications.
- Develop teamwork and collaboration skills.
- Strengthen problem-solving and critical thinking abilities by addressing real-world transportation challenges.
- Enhance communication and documentation skills through proposal writing, project reporting, and project presentation.
- Gain experience in project planning, time management, and agile development workflows.

1.3.4 Scope

The scope of the سُبل KSU project defines the boundaries of a web-based transportation management system designed to organize bus services managed by King Saud University only for female students, providing transportation between the metro station and the university female campus. supported by a centralized database to manage bus information, reservations, and real-time seat availability.
The system supports two main user roles: Administrator and Students
students can register and log in to search for available buses, view realtime seat availability and reserve seats or cancel reservations when needed, in addition they can allocate bus arrival times on a daily, weekly, or monthly basis to help them plan ahead and reduce congestion. Administrators manage system data by adding and updating bus records, managing bus availability and marking buses as out of service which automatically will be hidden in user views. The system will support the English language only and will be accessible through chrome, edge and safari.The scope of the project does not include mobile applications, additional language support or payment and real time GPS tracking or driver side interfaces. Any features not included in this scope are outside the current project and may be considered in future phases.

1.4 Scrum Team

Scrum Team	
Product Owner:	Razan Alkahtani
Developers:	• Dana Alotaibi • Gharsah Almusaeed • Razan Alkahtani • Ghena Binjadid
Scrum Master (SM):	Ms. Ghaida Alfayez
Stakeholders:	Ms. Ghaida Alfayez

References

[1] Google, "Gemini," 2026.